

Exercice 1

1) calculons Q_f

- à l'équilibre $Q_1 + Q_2 = 0$

$$\Rightarrow m_1 C_1 (\theta_f - \theta_1) + m_2 C_2 (\theta_f - \theta_2)$$

$$\Rightarrow \theta_f = \frac{m_1 C_1 \theta_1 + m_2 C_2 \theta_2}{m_1 + m_2}$$

$$\theta_f = \frac{60}{3} \quad \boxed{\theta_f = 20^\circ \text{C}}$$

2) calculons la quantité de chaleur nécessaire pour chauffer le mélange à 40 °C en joules

$$Q_1 = m C (\theta_f - \theta_i)$$

$$\text{AN: } Q_1 = 3 \cdot 10^3 \times 4,18 (40 - 20)$$

$$\boxed{Q_1 = 250800 \text{ J}}$$

x en calorie

$$Q_1 = 4,18 \text{ J} \rightarrow 1 \text{ cal}$$

$$250800 \text{ J} \rightarrow ?$$

$$\boxed{Q_1 = 6 \cdot 10^4 \text{ cal}}$$

3) calculons T_f du mélange

$$m_e = 3 \text{ kg}$$

$$\theta_e = 40^\circ \text{C}$$

$$C_e = 4,18$$

$$Q_c + Q_{Fe} = 0$$

$$\left. \begin{array}{l} m_{Fe} = 200 \text{ g} \\ \theta_{Fe} = 300^\circ \text{C} \end{array} \right| T_f$$

$$C_{Fe} = 20^\circ \text{C}$$

$$m_e C_e (T_f - \theta_e) + m_{Fe} C_{Fe} (T_f - \theta_{Fe}) = 0$$

$$T_f = \frac{m_e C_e \theta_e + m_{Fe} C_{Fe} \theta_{Fe}}{m_e C_e + m_{Fe} C_{Fe}}$$

$$\boxed{T_f = 102,88^\circ \text{C}}$$

Exercice 2

$$m = 56 \text{ g}; \quad p_1 = 1 \text{ atm}; \quad p_2 = 20 \text{ atm}$$

$$T = 25^\circ \text{C}; \quad M(\text{N}_2) \text{ diatomique} = 28$$

1) compression effectuée de manière réversible

$$\int_{V_1}^{V_2} -p dV \Rightarrow W = \int_{V_1}^{V_2} -p dV$$

N_2 est un gaz parfait

$$W = - \int_{V_1}^{V_2} \frac{nRT}{V} dV; \quad T = \text{cte} \Rightarrow$$

$$W = -nRT \int_{V_1}^{V_2} \frac{1}{V} dV$$

$$\Rightarrow W = -nRT \ln \left(\frac{V_2}{V_1} \right)$$

$$W = -nRT (\ln(V_2) - \ln(V_1))$$

x volume de V_1 et V_2

$$p_1 V_1 = nRT_1; \quad p_2 V_2 = nRT_2$$

$$T_1 = T_2 \Rightarrow$$

$$p_1 V_1 = p_2 V_2 \Rightarrow \frac{V_2}{V_1} = \frac{p_1}{p_2}$$

Exercice 4

$V_1 = 8,85 \text{ cm}^3$; $P_1 = 0,556 \text{ atm}$
 $T_1 = 30^\circ\text{C}$; $V_2 = 1000 \text{ cm}^3$
 $T_2 = 20^\circ\text{C}$

calculer la nouvelle

pression

$$P_1 V_1 = n R T_1 \quad ; \quad P_2 V_2 = n R T_2$$

$$\frac{P_1 V_1}{P_2 V_2} = \frac{n R T_1}{n R T_2}$$

$$P_2 V_2 = \frac{P_1 V_1 T_2}{T_1}$$

$$P_2 = \frac{P_1 V_1 T_2}{V_2 T_1}$$

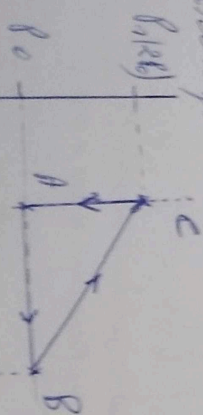
$$P_2 = \frac{0,556 \times 8,85 \times (20 + 273)}{1000 \times (30 + 273)}$$

$$P_2 = 0,44 \text{ atm}$$

Exercice 5

$P_0 = 2 \cdot 10^5 \text{ Pa}$; $V_0 = 14 \text{ L}$

2) Représenter le cycle des transformations dans un diagramme



Calculer la température dans le processus de compression isotherme

$$\text{en } D \quad P_0 V_0 = n R T_0 = R T_D$$

$$\text{en } B \quad P_0 (2V_0) = n R T_1 = R T_1$$

$$T_1 = \frac{2 P_0 V_0}{R}$$

$$P_1 V_1 = \frac{2 \times 2 \cdot 10^5 \times 14 \cdot 10^{-3}}{8,32}$$

$$T_1 = 673,07 \text{ K}$$

Calculer la pression maximale en C

$$P_1 V_0 = n R T_1 \Rightarrow P_1 = \frac{n R T_1}{V_0}$$

$$P_1 = \frac{R}{V_0} \cdot \frac{2 P_0 V_0}{R} \Rightarrow P_1 = 2 P_0$$

$$P_1 = 2 \times 2 \cdot 10^5 \text{ Pa}$$

3) Calculer le travail d'échange pour le système durant ce cycle

$$W_{\text{cycle}} = W_{AB} + W_{BC} + W_{CA}$$

$$W_{CA} = 0 \quad (\text{ou isochore})$$

$$W_{AB} = - \int_{V_0}^{2V_0} P_0 dV = - P_0 \int_{V_0}^{2V_0} dV$$

$$W_{AB} = - P_0 (2V_0 - V_0) = - P_0 V_0$$

exercice 5

$$W_{BC} = - \int_{V_B}^{V_C} P dV = - nRT_1 \int_{V_B}^{V_C} \frac{1}{V} dV$$

$$= - nRT_1 \ln\left(\frac{V_2}{2V_0}\right)$$

$$= - nR \cdot \frac{2P_0V_0}{R} \ln\left(\frac{1}{2}\right)$$

en remplaçant T_1 par son expression

$$T_1 = \frac{2P_0V_0}{R}$$

$$W_{BC} = - R \frac{2P_0V_0}{R} \ln\left(\frac{1}{2}\right)$$

$$W_B = 2P_0V_0 \ln 2$$

$$W_{\text{cycle}} = -P_0V_0 + 2P_0V_0 \ln 2 + 0$$

$$W_{\text{cy}} = (-1 + 2 \ln 2) P_0V_0$$

$W_{\text{cy}} > 0$ cycle récepteur

$$W = 1081,62 \text{ J}$$

- déterminons la quantité de chaleur échangée durant le système

$$\Delta U = W + Q$$

$$\Delta U_{\text{cycle}} = C_V \Delta T = \frac{nR}{\gamma - 1} (T_A - T_A)$$

$$\Rightarrow \Delta U_{\text{c}} = 0 \Rightarrow W + Q = 0$$

$$W = -Q = -W$$

$$Q = -1081,62 \text{ J}$$

donc $Q < 0$ donc le système cède de la chaleur

Exercice 6

$$V_1 = 2500 \text{ L}; T_1 = 20^\circ \text{C}$$

$$P_1 = 2 \text{ MPa} = 2 \cdot 10^6 \text{ Pa}$$

$$V_2 = 6000 \text{ L}; T_2 = 150^\circ \text{C}$$

La pression dans la machine

$$P_1 V_1 = nRT_1; P_2 V_2 = nRT_2$$

$$\frac{P_2 V_2}{P_1 V_1} = \frac{nRT_2}{nRT_1} = \frac{P_2 V_2}{P_1 V_1} = \frac{T_2}{T_1}$$

$$P_2 = \frac{P_1 V_1 T_2}{V_2 T_1}$$

$$\text{Avec: } P_2 = \frac{2 \cdot 10^6 \times 2500 \times 10^{-3} \times (150 + 273)}{6000 \cdot 10^{-3} \times (20 + 273)}$$

$$P_2 = 1,20 \cdot 10^6 \text{ Pa}$$

$$1 \text{ MPa} = 10^6 \text{ Pa}$$

$$1 \text{ l} \rightarrow 10^{-3} \text{ m}^3$$

$$6000 \text{ L} \rightarrow 6000 \cdot 10^{-3}$$

ent. exo S

$$W_{BC} = - \int_{V_B}^{V_C} P dV = - nRT_1 \int_{V_B}^{V_C} \frac{1}{V} dV$$

$$= - nRT_1 \ln\left(\frac{V_2}{2V_0}\right)$$

$$= - nR \cdot \frac{2P_0V_0}{R} \ln\left(\frac{1}{2}\right)$$

en remplaçant T_1 par
sa seule expression

$$T_1 = \frac{2P_0V_0}{R}$$

$$W_{BC} = - R \frac{2P_0V_0}{R} \ln\left(\frac{1}{2}\right)$$

$$W_B = 2P_0V_0 \ln 2$$

$$W_{\text{cycle}} = -P_0V_0 + 2P_0V_0 \ln 2 + 0$$

$$W_{\text{cy}} = (-1 + 2 \ln 2) P_0V_0$$

$W_{\text{cy}} > 0$ cycle récepteur

$$W = 1081,62 \text{ J}$$

- déterminons la quantité
de chaleur échangé durant
le système

$$\Delta U = W + Q$$

$$\Delta U_{\text{cycle}} = C_V \Delta T = \frac{nR}{\gamma-1} (T_A - T_A)$$

$$\Rightarrow \Delta U_{\text{c}} = 0 \Rightarrow W + Q = 0$$

$$W = -Q = -W$$

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La pression dans la machine

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$$\frac{P_2 V_2}{P_1 V_1} = \frac{nRT_2}{nRT_1} = \frac{P_2 V_2}{P_1 V_1} = \frac{T_2}{T_1}$$

$$P_2 = \frac{P_1 V_1 T_2}{V_2 T_1}$$

$$\text{A.N.: } P_2 = \frac{2 \cdot 10^6 \times 2500 \times 10^{-3} \times (150 + 273)}{6000 \times 10^{-3} \times (20 + 273)}$$

$$P_2 = 1,20 \cdot 10^6 \text{ Pa}$$

$$1 \text{ MPa} = 10^6 \text{ Pa}$$

$$1 \text{ l} \rightarrow 10^{-3} \text{ m}^3$$

$$6000 \text{ L} \rightarrow 6000 \cdot 10^{-3}$$

1. Suite Exo 9

$$\Delta H = C_p \Delta T \text{ (2e loi de Joule)}$$

$$= \frac{\gamma m R}{\gamma - 1} (T_2 - T_1)$$

$$= \frac{\gamma (m R T_2 - m R T_1)}{\gamma - 1}$$

$$\boxed{\Delta H = \frac{\gamma (P_2 V_2 - P_1 V_1)}{\gamma - 1}}$$

2) Montrons $du = \delta w$

$$du = \delta w + \delta q \text{ (forme diff)}$$

du 1er principe

or $\delta q = 0$ (car transformation adiabatique) $\Rightarrow \ln \left(\frac{T}{V^{\gamma-1}} \right) = \text{cte}$

$$\boxed{du = \delta w}$$

x La relation entre dT , dV , V , T , et γ ; on sait que

$$\delta w = du = C_v dT$$

$$\text{et } \delta w = -P dV \Rightarrow$$

$$C_v dT = -P dV$$

$$\frac{m R dT}{\gamma - 1} = - \frac{m R T dV}{V}$$

$$\frac{dT}{T} = - \frac{dV}{V}$$

$$\frac{dT}{T} = (\gamma - 1) \frac{dV}{V}$$

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$$3) \text{ Montrons } T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\frac{dT}{T} = (\gamma - 1) \frac{dV}{V}$$

$$\int \frac{dT}{T} = (\gamma - 1) \int \frac{dV}{V}$$

$$\ln T = (\gamma - 1) \ln V + \text{cte}$$

$$\ln T = \ln(V^{\gamma-1}) + \text{cte}$$

$$\ln T - \ln(V^{\gamma-1}) = \text{cte}$$

$$\ln \left(\frac{T}{V^{\gamma-1}} \right) = \text{cte}$$

$$T V^{\gamma-1} = \text{cte}$$

$$\boxed{T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}}$$

En déduisons

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$T V^{\gamma-1} = \text{cte}$$

$$P V = n R T \text{ (car gaz parfait)}$$

$$T = \frac{P V}{n R} \Rightarrow \frac{P V}{n R} V^{\gamma-1} = \text{cte}$$

$$P V^\gamma = n R \text{ cte} \Rightarrow P V^\gamma = \text{cte} \Rightarrow$$

$$\boxed{P_1 V_1^\gamma = P_2 V_2^\gamma}$$

3) déterminons P_1

$$\gamma = 1,4; V_1 = \frac{V_0}{10}; P_0 = 1 \text{ bar}$$

$$P_1 = ?; V_0 = ?$$

utilisons la loi de la place

$$P_1 V_1^\gamma = P_0 V_0^\gamma \Rightarrow P_1 = \frac{P_0 V_0^\gamma}{V_1^\gamma}$$

$$P_1 = \frac{P_0 V_0^\gamma}{\left(\frac{V_0}{10}\right)^\gamma} \Rightarrow P_1 = \frac{P_0 V_0^\gamma 10^\gamma}{V_0^\gamma}$$

$$P_1 = 10^\gamma P_0 \Rightarrow P_1 = 10^{1,4} \times 1$$

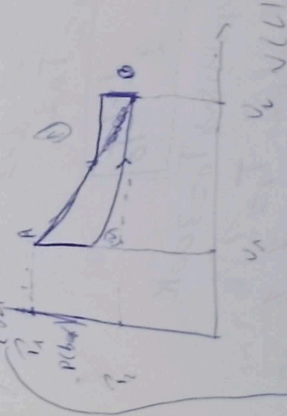
$$\boxed{P_1 = 25,12 \text{ bar}}$$

Exercice 11

$$m = 0,1 \text{ mol} \quad \left| \begin{array}{l} P_1 = \\ V_1 = 5,0 \text{ L} \\ T_1 = 287 \end{array} \right. \quad \left| \begin{array}{l} P_2 = \\ V_2 = 20 \\ T_2 = 350 \text{ K} \end{array} \right.$$

1) Représentons ces états sur le

diagramme



de l'exercice 10

$$P_1 V_1 = n R T_1$$

$$P_1 = \frac{n R T_1}{V_1} = \frac{0,1 \times 8,32 \times 287}{5,0 \cdot 10^{-3}}$$

$$P_1 = 2384,97 \text{ Pa} = 2,38 \text{ bar}$$

$$P_2 V_2 = n R T_2$$

$$P_2 = \frac{n R T_2}{V_2} = \frac{0,1 \times 8,32 \times 350}{20 \cdot 10^{-3}}$$

$$\boxed{P_2 = 727,12 \text{ Pa} = 0,72 \text{ bar}}$$

2) Écrivons pr. chaque p. le

ws. chimico, le bilan

énergétique total ($\Delta U, Q$

& chemins)

$$\Delta U = C_V \Delta T \text{ or } C_V = \frac{nR}{\gamma - 1}$$

$$\Delta U = \frac{nR}{\gamma - 1} (T_2 - T_1) = \frac{0,1 \times 8,32}{1,4 - 1} \times (350 - 287)$$

$$\boxed{\Delta U = 654,41 \text{ J}}$$

* le travail W

$$W_{AB} = W_{CA} + W_{CB} \text{ or}$$

$$W_{CB} = 0 \text{ car isochore}$$

$$W_{AB} = W_{AC}$$

$$W =$$